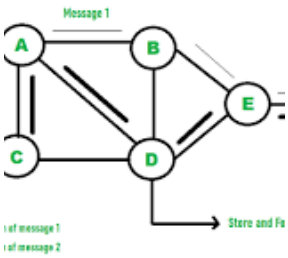
A 2.3.3 Packet Switching



A2.3.3 Explain how packet switching is used to send data across a network.

Packet switching involves segmenting data into packets with routing headers attached. These packets are then transmitted independently across the network. At the destination, the data is reassembled

Introduction

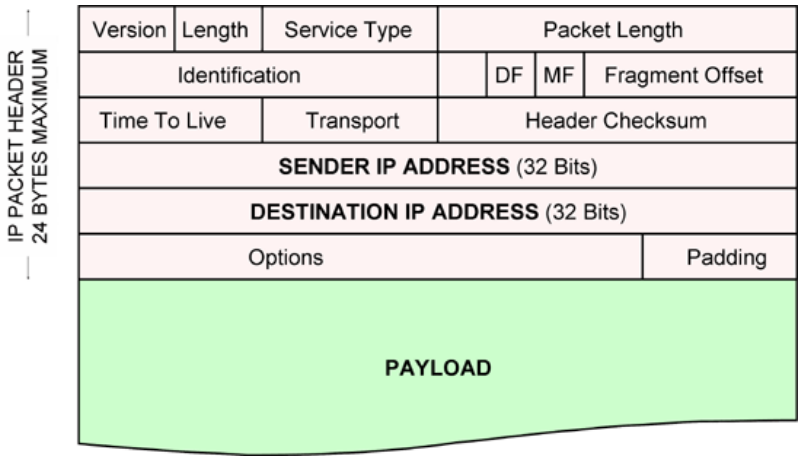
Imagine sending a letter. Instead of sending it as a single sheet, you cut it into smaller pieces, put each piece in an envelope with the recipient's address, and send them separately. They might take different routes, but they all eventually reach the destination, where they are reassembled into the original letter. This is the essence of packet switching, a fundamental technique for transmitting data across networks.

1. Segmenting Data and Independent Transmission

In packet switching, data is divided into small units called packets. Each packet contains:

Data: A portion of the original message, also known as the **payload**.
Header: Control information, including the source and destination IP addresses, packet sequence number, and error-checking codes.

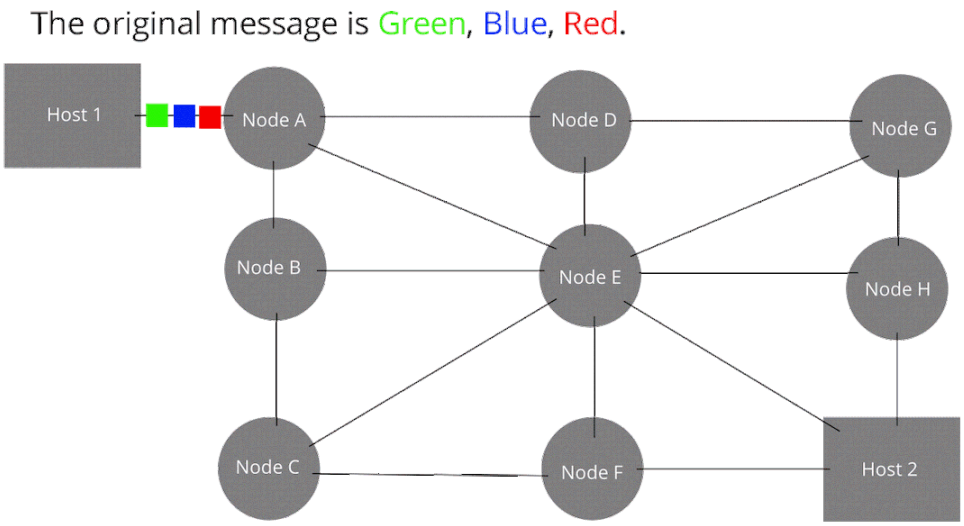
These packets are then transmitted independently across the network. Depending on network conditions and traffic, they may take different routes to reach the destination. This flexibility and efficiency are key advantages of packet switching.



IPV4 data packet header.

2. Reassembly at the Destination

Once all the packets arrive at the destination, they are reassembled in the correct order based on their sequence numbers. The header information ensures that all packets have arrived and are in order. If any packets are missing or corrupted, they can be retransmitted.



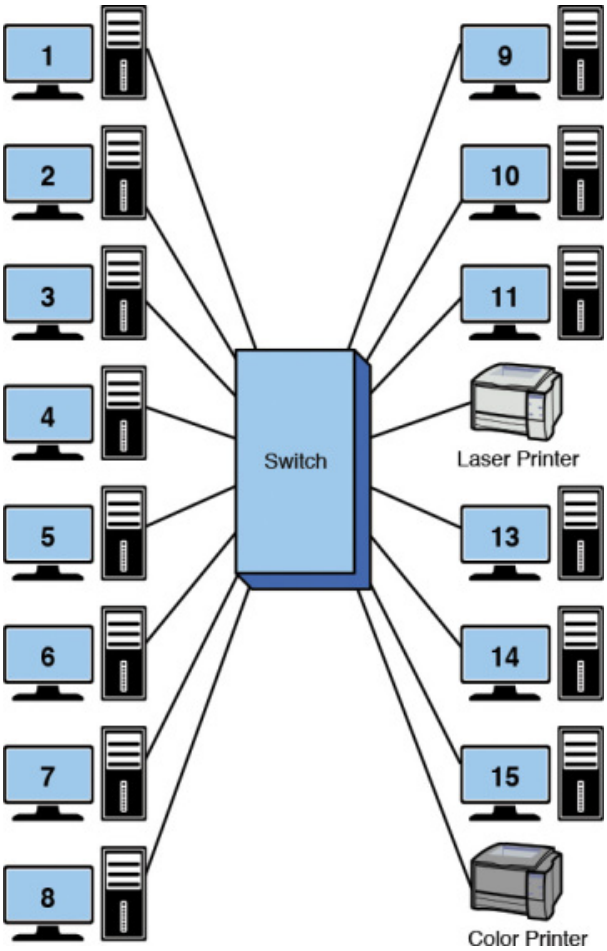
A diagram showing packets arriving at the destination and being reassembled into the original message.

3. The Role of Switches and Routers

Switches:

Switches connect devices within a local area network (LAN). They use MAC addresses to forward packets to their correct destination within the network.

How they work: Switches learn the MAC addresses of devices connected to them and store them in a table. When a packet arrives, the switch examines the destination MAC address and forwards the packet to the corresponding port.

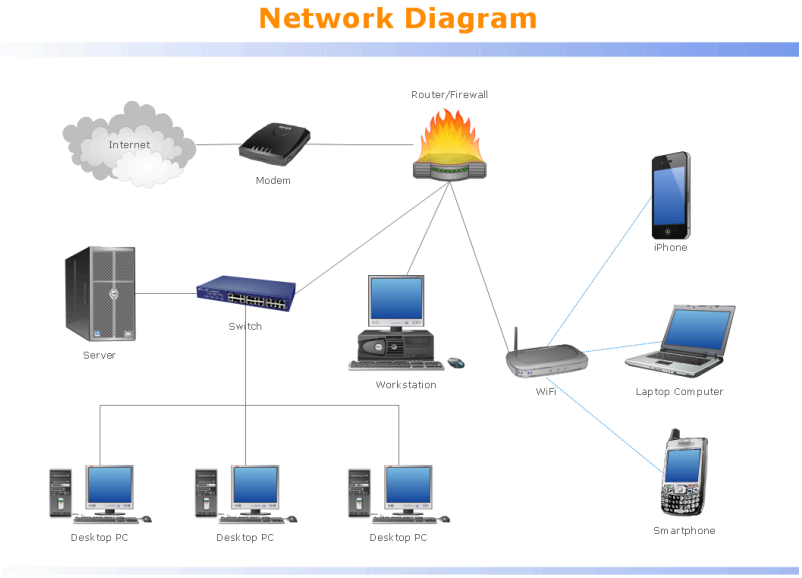


A diagram showing a switch connecting multiple devices in a LAN.

Routers:

Routers connect different networks together, such as a LAN to the internet. They use IP addresses to forward packets between networks.

How they work: Routers examine the destination IP address in a packet's header and determine the best route to forward the packet towards its destination. They maintain routing tables that store information about network paths.



A diagram showing a router connecting a LAN to the internet.

Key Questions

Q: What are the advantages of packet switching over other data transmission methods?

Q: How does the header information in a packet ensure reliable data delivery?

Q: What is the difference between a switch and a router?

Q: How do routers determine the best path to forward a packet?

Q: What happens if a packet is lost or corrupted during transmission?

Q: Can you explain how packet switching enables different data types (e.g., text, images, video) to be transmitted over the same network?

Q: Why is packet switching crucial for the functioning of the internet?